

Note on Solid State Theory

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Spring 2026

Abstract

This is the note for David Tong's lecture note on *Quantum Hall Effect*. And I also would like to include lecture notes taught by Yu-An Chen on **Solid State Theory** in the future.

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1 An Aside: Quantum Computing

Some Conventions

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

The spin lowering and raising operators are defined by

$$\sigma^- = \frac{1}{2}(X - iY) \quad \sigma^+ = \frac{1}{2}(X + iY)$$

1.1 Quantum Error Correction

Parent Hamiltonian

Parent Hamiltonian defines an energy framework that ground state represents error-free states. In the case of a three-qubit repetition code, the parent Hamiltonian is

$$H = -J(Z_1 Z_2 + Z_2 Z_3) \tag{1.1}$$

where $Z_1 Z_2$ and $Z_2 Z_3$ are called *stabilizers*.

Ground states

There is a two-fold degeneracy of ground states, denoted as $|0\rangle = |000\rangle$ and $|1\rangle = |111\rangle$. Any derivation of ground states would increase the energy of this system.

Logical operators

Logical operators are defined such that they are non-trivial map on the code space. They are defined by modulo out stabilizers, which implies if two logical operators differ from each other only by a multiplication of stabilizers, they are equivalent. In our case,

$$\begin{aligned} \bar{X} &= X_1 X_2 X_3 : \bar{X} |0\rangle = |1\rangle \quad \bar{X} |1\rangle = |0\rangle \\ \bar{Z} &= Z_i : \bar{Z} |0\rangle = |0\rangle \quad \bar{Z} |1\rangle = -|1\rangle \end{aligned} \tag{1.2}$$

A brief summary

Stabilizers are defined to construct Hamiltonian, while logical operators are those commute with stabilizers but acts non-trivially on code space. The minimal Pauli weight of logical operators is called *code distance* of a quantum code.

2 1D topological phenomena

We study two mathematically equivalent models, *transverse field Ising model* and *Kitaev chain*. It can be shown using *Jordan-Wigner transformation* the two models can be mapped from one to the other.

2.1 AKLT Model

AKLT model, named after Affleck-Kennedy-Lieb-Tasaki, is defined by following Hamiltonian

$$H = \sum_i \left[\vec{S}_i \cdot \vec{S}_{i+1} + \frac{1}{3} (\vec{S}_i \cdot \vec{S}_{i+1})^2 \right] \quad (2.1)$$

The exact form could be thought as any two adjacent spin-1. There is a non-local transform

3 Berry Curvature

Consider a Hamiltonian with parameters varying slowly with time, $\vec{R}(t)$. Those parameters could be external magnetic fields, electric fields and so on. Since the system is changed adiabatically, we assume that an eigenstate would remain to be instantaneous eigenstate as parameters running up to a phase factor. We define the energy eigenstate in the parameter space as

$$H(\vec{R}) |n(\vec{R})\rangle = E_n(\vec{R}) |n(\vec{R})\rangle \quad (3.1)$$

Therefore we write the evolving state as

$$|\psi(t)\rangle = e^{-i\theta(t)} |n(\vec{R}(t))\rangle \quad (3.2)$$

Compute the Schrödinger equation with adiabatic ansatz, we find the differential equation satisfied by phase factor,

$$E_n(\mathbf{R}(t)) - i\hbar \langle n(\mathbf{R}(t)) | \frac{d}{dt} | n(\mathbf{R}(t)) \rangle = \hbar \frac{d\theta(t)}{dt}. \quad (3.3)$$

Integrate with time, we find there are two different contribution, one comes entirely from dynamical evolution, the other arises from the geometry of parameter space. We are mostly interested in the second one, which is so-called *Berry phase*.

$$\gamma_n = i \int_0^t dt' \left\langle n(\vec{R}(t')) \left| \frac{d}{dt'} \right| n(\vec{R}(t')) \right\rangle \quad (3.4)$$

This motivates the definition of *Berry connection*, which is defined as a one-form on the parameter space of the quantum system.

$$\mathbf{A}_n(\mathbf{R}) = i \langle n(\mathbf{R}) | \nabla_{\mathbf{R}} | n(\mathbf{R}) \rangle \quad (3.5)$$

Clearly, this is not gauge-invariant, where we have defined the gauge transform as the arbitrary phase choice of the eigenstates, and the story is the same as we encounter in the electromagnetism. One could show that the Berry phase could be written in terms of Berry connection in a compact form,

$$\gamma_n = \oint_{\mathcal{C}} \mathbf{A}_n(\mathbf{R}) \cdot d\mathbf{R} \quad (3.6)$$

where we have assumed that we return to the same point in the parameter space. By Stoke's theorem, we may define *Berry curvature* if the space is three-dimensional, which

could also be thought as the field strength, given that Berry connection being a gauge potential.

A further computation shows that Berry phase could be put into a computationally convenient form, that is, we may compute it with derivatives of Hamiltonian in stead of derivatives of states.

$$\gamma_n = - \int_S d\mathbf{S} \cdot \mathbf{V}_n \quad (3.7)$$

where $\mathbf{V}_n$ is defined as

$$\mathbf{V}_n = \text{Im} \sum_{m \neq n} \frac{\langle n(\mathbf{R}) | (\nabla_R H) | n(\mathbf{R}) \rangle \langle n(\mathbf{R}) | (\nabla_R H) | m(\mathbf{R}) \rangle}{(E_m(\mathbf{R}) - E_n(\mathbf{R}))^2} \quad (3.8)$$

that is related with Berry curvature via $V_{ni} = \epsilon_{ijk} F_{jk}$.

A two-level system

Consider following Hamiltonian

$$H = \mathbf{d}(\mathbf{R}) \cdot \boldsymbol{\sigma} \quad (3.9)$$

we have ignored contribution from identity matrix as it only shifts enegies by a constant. We parametrized $\mathbf{d}$ in spherical coordinate

$$\mathbf{d}(\mathbf{R}) = |d|(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \quad (3.10)$$

A convenient choice of eigenstates with energies $\pm|d|$ is

$$|+\mathbf{R}\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix} \quad |-\mathbf{R}\rangle = \begin{pmatrix} \sin \frac{\theta}{2} e^{-i\phi} \\ -\cos \frac{\theta}{2} \end{pmatrix} \quad (3.11)$$

However, such choice of eigenstates is not well-defined globally. That is, at the south pole where $\theta = \pi$, eigenstates may vary as ϕ changes, indicating a singularity. This would not make any problme as we know from differential geometry this can be resolved by gluing different patches on the two-sphere. The Berry curvature is easy to compute and one can show it takes the same form even we change from one patch to the other,

$$F_{\theta\phi} = \frac{1}{2} \sin \theta. \quad (3.12)$$

Generally, if we take the Cartesian coordinate system and choose $\mathbf{d} = \mathbf{R}$. Then $H = x\sigma_x + y\sigma_y + z\sigma_z$. Assume that at a specific time, $\mathbf{R}$ is along z-direction and choose eigenstates to be the usual ones, satisfying

$$\sigma_z |+\rangle = |+\rangle \quad \sigma_z |-\rangle = -|-\rangle \quad (3.13)$$

Then $\mathbf{V}_+$ is computed by

4 The Integer Quantum Hall Effect

In this section, we would like to discuss *the integer quantum Hall effect* first.

Laudau levels

[Briefly discuss Laudau levels](#)

4.1 General arguments

Quantum Hall effect is that one finds that a material's Hall resistivity takes special value

$$\rho_{xy} = \frac{2\pi\hbar}{e^2} \frac{1}{\nu} \quad \nu \in \mathbb{Z} \quad (4.1)$$

If we compare this result with the classical calculation in Drude model, in which the resistivity is found to be,

$$\rho_{xy} = \frac{B}{ne} \quad (4.2)$$

where B is the external magnetic field and n is the electron density, one readily finds that on the ν -th plateau,

$$n = \frac{B}{\Phi_0} \nu \quad (4.3)$$

where $\Phi_0 = \frac{2\pi\hbar}{e}$ is the flux quantum in the usual sense. Let us first make a comment on this calculation.

Indeed, this is value of magnetic field we find on the center of the plateau, however not enough to explain the existence of plateau. What would happen when we move away from the specific value of magnetic field? Given that the material is pure, there is no where for electrons to go if we assume $k_B T \ll \hbar\omega_B$, where $\hbar\omega_B$ is the energy gap between two different Laudau levels. Actually, this is a result of impurity of materials, thus somewhere *disorder* should be taken into account.

The role of disorder, part I It is quite easy to see why disorder may help us to solve this problem. Mathematically, we introduce a random potential to the Hamiltonian, which effectively model the effect of disorder. We further require that the impurity is not that much, so that the typical value of the random potential is much smaller compared with the gap,

$$V \ll \hbar\omega_B. \quad (4.4)$$

Intuitively, this potential would not change the behaviour of Laudau level significantly, that is, the center of each Laudau level would not shift, at least the leading order of such perturbation. However, degenerate states would split due to such general perturbation, since they do not preserve any symmetry of original Hamiltonian. See fig 1 for an illustration.

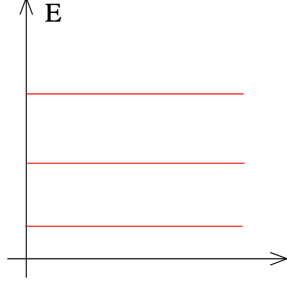


Figure 16: Density of states without disorder...

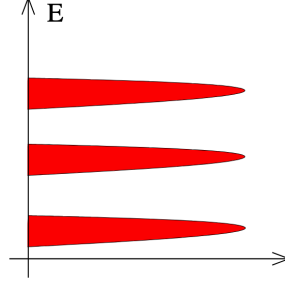


Figure 17: ...and with disorder.

Figure 1: States in the presence of disorder

The role of disorder, part II On the other hand, such random potential may take *extended states* into *localized states*, since impurity would prevent a electron from moving far away from its original position. We can easily see the emergence of localized states at the classical level. We assume that

$$|\nabla V| \ll \frac{\hbar\omega_B}{l_B} \quad (4.5)$$

where l_B is the magnetic radius. Recall that if we use the symmetric gauge, states could be labelled by the center of the orbit (X, Y) . One could again compute the commutator of them with the full Hamiltonian,

$$\begin{aligned} i\hbar\dot{X} &= [X, H + V] = [X, V] = [X, y] \frac{\partial V}{\partial y} = il_B^2 \frac{\partial V}{\partial y} \\ i\hbar\dot{Y} &= [Y, H + V] = [Y, V] = [Y, x] \frac{\partial V}{\partial x} = -il_B^2 \frac{\partial V}{\partial x} \end{aligned} \quad (4.6)$$

which means the center would move along the direction of equipotentials. So why this means we have localized states and extended states? First, it is easily to see states are extended without random potential as one can solve the wave function for each Landau level and in the Landau gauge the translation symmetry along y -direction means they are extended along this direction. Second, in the existence of random potential we may assume there are local maximums and local minimums. States near those extremums could only move near them, as they have been trapped close to extremums due to the force of random potential. And states that have medium energy could move along equipotentials that expand largely in the space, thus being referred to extended states. See fig 2 for an illustration.

Intuitively, only extended states can transport charge from one side of material to the other. Then as we fill all the extended states in a given Landau level and decrease the magnetic field B , each Landau level can accomodate fewer states. However rather than jump to the next Landau level directly, states change from extended ones to localized ones. That is, we have a plateau before localised states are also fully filled and we may jump to the next plateau.

4.2 Topology associated with quantum Hall effect

In this subsection, we would like to relate the quantum Hall effect with topology.

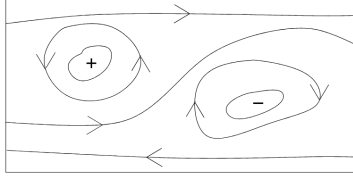


Figure 18: The localisation of states due to disorder.

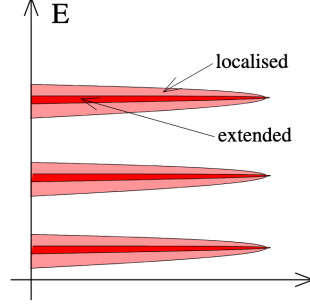


Figure 19: The resulting density of states.

Figure 2: Localized states and extended states

The first one we would like to derive is the *Kubo formula*, which relates the Hall conductivity with quantum observables and is part of *linear response*.

Consider a general Hamiltonian H_0 with energy eigenstates $H_0 |n\rangle = E_n |n\rangle$. We would like to add a background electric field and work in the gauge $A_0 = 0$. The electric field is $\mathbf{E} = -\partial_t \mathbf{A}$ and electrons are coupled with this by adding an extra term $\Delta H = -\mathbf{J} \cdot \mathbf{A}$ to the Hamiltonian. We would like to consider DC field however work with an oscillating field. That is we take $\mathbf{E}(t) = \mathbf{E}e^{-i\omega t}$ first and take $\omega \rightarrow 0$ at the end of our calculation. For such a electric field, the gauge potential is given by $\mathbf{A} = \frac{\mathbf{E}}{i\omega}e^{-i\omega t}$. Our goal is to compute the current $\langle \mathbf{J}(t) \rangle$ induced in this circumstance.

We would like to work in the interacting picture, where operators evolve as $\mathcal{O}(t) = e^{iH_0 t/\hbar} \mathcal{O}(0) e^{-iH_0 t/\hbar}$ and states evolve by

$$|\psi(t)\rangle_I = U(t, t_0) |\psi(t_0)\rangle_I \quad (4.7)$$

where the evolution operator is defined as

$$U(t, t_0) = \text{T exp} \left(-\frac{i}{\hbar} \int_{t_0}^t dt' \Delta H(t') \right) \quad (4.8)$$

where T stands for the time ordering. The computation is straightforward. Suppose that we prepare our state in the many-body ground state $|0\rangle$ in the far-past,

$$\begin{aligned} \langle \mathbf{J}(t) \rangle &= \langle 0(t) | \mathbf{J}(t) | 0(t) \rangle \\ &= \langle 0 | U^{-1}(t) \mathbf{J}(t) U(t) | 0 \rangle \\ &\approx \langle 0 | \mathbf{J}(t) + \frac{i}{\hbar} \int_{-\infty}^t dt' [\Delta H(t'), \mathbf{J}(t)] | 0 \rangle \end{aligned} \quad (4.9)$$

where $U(t)$ is the abbreviation of evolution operator from past-infinity to t and we have already expand $U(t)$ to leading order in the last step. Each component can be shown to be brought to following form

$$\langle J_i(t) \rangle = -\frac{1}{\hbar\omega} \left(\int_0^\infty dt'' e^{i\omega t''} \langle 0 | [J_j(0), J_i(t'')] | 0 \rangle \right) E_j e^{-i\omega t}. \quad (4.10)$$

From the above formula, it is clear that the diagonal component of conductivity is 0 and the off-diagonal components is

$$\sigma_{xy}(\omega) = -\frac{1}{\hbar\omega} \left(\int_0^\infty dt e^{i\omega t} \langle 0 | [J_j(0), J_i(t)] | 0 \rangle \right). \quad (4.11)$$

This is the *Kubo formula* of Hall conductivity. Use the evolution of current operator we mentioned above, this could be expanded using the eigenstates of H_0 and perform the integral over $\int dt$, where an extra small imaginary part should be introduced to make the integral convergent, $\omega \rightarrow \omega + i\epsilon$. The final result is

$$\sigma_{xy}(\omega) = \frac{i}{\omega} \sum_{n \neq 0} \frac{\langle 0 | J_y | n \rangle \langle n | J_x | 0 \rangle}{E_n - E_0 + \hbar\omega} - \frac{\langle 0 | J_x | n \rangle \langle n | J_y | 0 \rangle}{\hbar\omega - E_n + E_0}. \quad (4.12)$$

Expand around $\omega \rightarrow 0$ and throw away the divergent part (which would be important for the longitudinal conductivity however we are not going to discuss it here), we have

$$\sigma_{xy} = -i\hbar \sum_{n \neq 0} \frac{\langle 0 | J_y | n \rangle \langle n | J_x | 0 \rangle - \langle 0 | J_x | n \rangle \langle n | J_y | 0 \rangle}{(E_n - E_0)^2} \quad (4.13)$$

Now we are ready to discuss the topological nature of quantum Hall effect. Let us consider the Hall system on a torus $\mathbf{T}^2$, which can be viewed as a rectangle with opposite sites identified. First, we can no longer impose periodic boundary condition for electrons due to the existence of magnetic field. We introduce the *magnetic translation operators*,

$$T(\mathbf{d}) = \exp(i\mathbf{d} \cdot (-i\nabla - e\mathbf{A}/\hbar)). \quad (4.14)$$

We require that the state is invariant under the translation of the torus by the magnetic translation operators. However, it is clear that the wave function would acquire a phase when we translate it around a circle of the torus, which is also gauge-dependent. If we take the Landau gauge $A_x = 0, A_y = Bx$, then T_x is the usual translation operator but $T_y = \exp(y\partial_y - iyeBx/\hbar)$. That means when we move around a circle along y -direction, the wave function would pick up a phase

$$T(0, L_y)\psi(x, y) = \exp(-ieBL_y/\hbar)\psi(x, y + L_y) = \psi(x, y). \quad (4.15)$$

However, we find that $T_x T_y = \exp(-ieBL_x L_y/\hbar) T_y T_x$. That is, the two translation operators do not commute with each other. To make them commute, we require the factor to be a multiple of 2π ,

$$\frac{eBL_x L_y}{\hbar} = 2\pi N \quad N \in \mathbb{Z}, \quad (4.16)$$

which give rise to the Dirac quantization condition for magnetic flux. It is again easy to see that we could perturb the Hamiltonian by adding constant magnetic flux along two directions.

$$\begin{aligned} A_x &= \frac{\Phi_x}{L_x} \\ A_y &= \frac{\Phi_y}{L_y} + Bx \end{aligned} \quad (4.17)$$

The addition Hamiltonian is $\Delta = - \sum_{i=x,y} \frac{J_i \Phi_i}{L_i}$. Then to the first order in the perturbation theory, the energy eigenstates are given by

$$|\psi_0'\rangle = |\psi_0\rangle + \sum_{n \neq \psi_0} \frac{\langle n | \Delta H | \psi_0 \rangle}{E_n - E_0} |n\rangle \quad (4.18)$$

Consider infinitesimal changes of Φ_i , and we differentiate above formula with respect to Φ_i , it can be shown that the right-handed side takes the form of Kubo formula. In all, the Kubo formula for Hall conductivity can be written as

$$\sigma_{xy} = -i\hbar \left[\frac{\partial}{\partial \Phi_y} \left\langle \psi_0 \left| \frac{\partial \psi_0}{\partial \Phi_x} \right\rangle - \frac{\partial}{\partial \Phi_x} \left\langle \psi_0 \left| \frac{\partial \psi_0}{\partial \Phi_y} \right\rangle \right] \quad (4.19)$$

Again, we surprisingly find that it takes a very similar form with the Berry curvature in the parameter space! If we set $\theta_i = \frac{2\pi\Phi_i}{\Phi_0}$ being angular variables, which should take value in $[0, 2\pi)$ since the spectrum is invariant if the magnetic flux is a multiple of flux quanta, being *spectral flow* that we did not discuss here.

If we define the Berry connection which, in this case, lives over $\mathbf{T}_{\Phi}^2$, by

$$\mathcal{A}_i = -i \langle \psi_0 | \frac{\partial}{\partial \theta_i} | \psi_0 \rangle. \quad (4.20)$$

The field strength can be easily computed and can be shown to be directly related with Hall conductivity

$$\sigma_{xy} = -\frac{e^2}{\hbar} \mathcal{F}_{xy} \quad (4.21)$$

To see why the Hall conductivity is quantized, one needs to average over the parameter space

$$\sigma_{xy} = -\frac{e^2}{\hbar} \int_{\mathbf{T}_{\Phi}^2} \frac{d^2\theta}{(2\pi)^2} \mathcal{F}_{xy} \quad (4.22)$$

However an integral of curvature over the space is always a integer, which is called **Chern number**. Here the Chern number is defined by

$$C = -\frac{1}{2\pi} \int_{\mathbf{T}_{\Phi}^2} \frac{d^2\theta}{(2\pi)^2} \mathcal{F}_{xy} \quad \text{and} \quad C \in \mathbb{Z} \quad (4.23)$$

The Hall conductivity is thus quantized

$$\sigma_{xy} = \frac{e^2}{2\pi\hbar} C \quad (4.24)$$

The relation (4.24) is so-called *TKNN invariant*.

Comment: Why taking average here?

It seems hard to understand why we take an average over θ here for the first time one encounter this formula. However the physics is rather simple. Terms used to perturb Hamiltonian can be thought as twisted boundary conditions, which need to be averaged when we want physical observable conductivity.

Laughlin's argument

We briefly discuss Laughlin's argument for quantized Hall effect. We adopt Landau's gauge

$$A = (0, Bx, 0)$$

In this gauge, the translation symmetry in y -direction is preserved and k_y is a good quantum number. It can be shown that the eigenstates of Hamiltonian is labeled by level

of oscillator in x -direction and k_y . As one may compute directly, k_y effectively shifts the center of oscillators in x -direction, $\langle x \rangle = -l^2 k_y$. Once we impose periodic boundary condition in y -direction, k_y is quantized

$$k_y = \frac{2\pi n}{L_y}$$

We also have a finite length in x -direction, say L_x , then degeneracy of a Landau level is given by

$$N = \frac{\Delta k_y}{\delta k_y} = \frac{L_x/l^2}{2\pi/L_y} = \frac{A}{2\pi l^2} = \frac{AB}{\Phi_0}$$

If we adiabatically increase flux in z -direction, which means $A_y = Bx + \frac{\Phi}{L_y}$, then

$$\langle x \rangle = -l^2(k_y + \frac{e\Phi}{\hbar L_y}) = -\frac{2\pi l^2}{L_y}(n + \frac{\Phi}{\Phi_0})$$

from which we find that if we add one unit of flux, the position of oscillator is shifted from n to $n + 1$, and the system doesn't change but each quantum state evolves to the next level. Thus if a Landau level is full filled, then such *spectral flow* would effectively translate one state from most inner orbit to an outer orbit, meaning one electron is moving. The Hall conductance can be read from following formula and $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$

$$Q = \int dt \oint j_r dl = \sigma_{xy} \int dt \oint E_\theta dl = -\sigma_{xy} \Delta \Phi$$

Aside: In this case if we label degenerate states in a given Landau level by n where $n = 0, 1, \dots, N - 1$, then it is weird at first sight where does the most outer state move. This is explained by that the $n = N$ state is an edge state, which is not completely degenerate with bulk states.

4.3 Topological insulators

So far so good, we have already explain the relation between quantized Hall conductivity and topology. In this section, we would like to discuss more practical models. Let us consider particles on a lattice. The first thing that one should be aware of is that now momentum should be restricted in *Brillouin zone*.

Definition 1 (Brillouin zone). *Consider that particles are living in 1d space, that is $\mathbb{R}$, the momentum eigenfunction is given by e^{ikx} and k would take value in $\mathbb{R}$. However if we restrict ourselves to 1d lattice with lattice distance a , such that particles have quantized position $x_n = na$, then one finds that for $\psi_k(x_n) = e^{ikna}$, this is invariant under $k \rightarrow k + G$, where $G = \frac{2\pi}{a}$. Thus one can not tell the difference between k and $k + G$, and the momentum space should modulo out such a translation. One may take*

$$k \in [-\frac{\pi}{a}, \frac{\pi}{a}) \quad (4.25)$$

as the first Brillouin zone.

Given the definition of Brillouin zone, we consider a particle living in a rectangular lattice, where the lattice distance are given by a and b in x - and y -direction, respectively.

The topology of momentum space is a torus $\mathbf{T}^2$ as we have the identification $k_x \sim k_x + \frac{2\pi}{a}$ and $k_y \sim k_y + \frac{2\pi}{b}$. In a given band, the wave function can be brought to the Bloch form

$$\psi_{\mathbf{k}}(\mathbf{x}) = e^{i\mathbf{k}\cdot\mathbf{x}} u_{\mathbf{k}}(\mathbf{x}) \quad (4.26)$$

where $u_{\mathbf{k}}(\mathbf{x})$ is usually periodic in a unit cell. We should make several assumptions before moving into concrete calculation

- First, we assume that the single particle spectrum decompose into bands, with each band parameterized by the momentum $\mathbf{k}$ which lives on a torus.
- Second, we assume that particles are non-interacting, and the multi-particle spectrum is constructed by filling up the single-particle spectrum, subject to Pauli exclusion principle.
- Finally, we assume that there is a energy gap between bands and there exist a Fermi energy E_F lives in one of these gaps. States below E_F are fully filled while bands above E_F are empty.

For each bands that below the Fermi energy, we can do the same thing as above. For each band there exist a Berry connection defined as

$$\mathcal{A}_i(\mathbf{k}) = -i \langle u_{\mathbf{k}} | \frac{\partial}{\partial k_i} | u_{\mathbf{k}} \rangle. \quad (4.27)$$

Here the connection is defined in the Hilber space itself, which is different from what we did before. Desipte of that, the story goes in the usual sense. The Berry curvature associate with $\mathcal{A}_i$ is

$$\mathcal{F}_{xy} = \frac{\partial \mathcal{A}_y}{\partial k^x} - \frac{\partial \mathcal{A}_x}{\partial k^y} = i \left\langle \frac{\partial u}{\partial k^y} \left| \frac{\partial u}{\partial k^x} \right. \right\rangle - i \left\langle \frac{\partial u}{\partial k^x} \left| \frac{\partial u}{\partial k^y} \right. \right\rangle \quad (4.28)$$

For each band α there is an associate Chern number and the Hall conductivity is given by a summation

$$\sigma_{xy} = \frac{e^2}{2\pi\hbar} \sum_{\alpha} C_{\alpha} \quad (4.29)$$

A brief proof of formula (4.29). The ground state is given by a tensor of single-particle states below the Fermi surface, schematically $|\Psi_0\rangle = \prod_{E_{\alpha} < E_F} c_{\alpha}(\mathbf{k})^{\dagger} |0\rangle$. An insertion of current operator between excited state and ground state effectively becomes $\langle u_{\alpha}^{\mathbf{k}} | J_y | u_{\beta}^{\mathbf{k}} \rangle$.

Let us look at the Schödinger equation of Bloch wave function, which reads

$$\begin{aligned} H |\psi_{\mathbf{k}}\rangle = E_{\mathbf{k}} |\psi_{\mathbf{k}}\rangle &\Rightarrow (e^{-i\mathbf{k}\cdot\mathbf{x}} H e^{i\mathbf{k}\cdot\mathbf{x}}) |u_{\mathbf{k}}\rangle = E_{\mathbf{k}} |u_{\mathbf{k}}\rangle \\ &\Rightarrow \tilde{H}(\mathbf{k}) |u_{\mathbf{k}}\rangle = E_{\mathbf{k}} |u_{\mathbf{k}}\rangle \quad \text{with} \quad \tilde{H}(\mathbf{k}) = e^{-i\mathbf{k}\cdot\mathbf{x}} H e^{i\mathbf{k}\cdot\mathbf{x}} \end{aligned} \quad (4.30)$$

The current operator is defined in terms of the group velocity $\mathbf{J} = \frac{e}{\hbar} \frac{\partial \tilde{H}}{\partial \mathbf{k}}$. One can check

that this perfectly match $\mathbf{J} = e\dot{\mathbf{x}}$. Now we are left with algebraric calculation

$$\begin{aligned}
\sigma_{xy} &= -i\hbar \sum_{E_\alpha < E_F < E_\beta} \int_{\mathbf{T}^2} \frac{d^2k}{(2\pi)^2} \frac{\langle u_\alpha^{\mathbf{k}} | J_y | u_\beta^{\mathbf{k}} \rangle \langle u_\beta^{\mathbf{k}} | J_x | u_\alpha^{\mathbf{k}} \rangle - \langle u_\alpha^{\mathbf{k}} | J_x | u_\beta^{\mathbf{k}} \rangle \langle u_\beta^{\mathbf{k}} | J_y | u_\alpha^{\mathbf{k}} \rangle}{(E_\alpha(\mathbf{k}) - E_\beta(\mathbf{k}))^2} \\
&= -i\frac{e^2}{\hbar} \sum_{E_\alpha < E_F < E_\beta} \int_{\mathbf{T}^2} \frac{d^2k}{(2\pi)^2} \frac{\langle u_\alpha^{\mathbf{k}} | \partial_y \tilde{H} | u_\beta^{\mathbf{k}} \rangle \langle u_\beta^{\mathbf{k}} | \partial_x \tilde{H} | u_\alpha^{\mathbf{k}} \rangle - \langle u_\alpha^{\mathbf{k}} | \partial_x \tilde{H} | u_\beta^{\mathbf{k}} \rangle \langle u_\beta^{\mathbf{k}} | \partial_y \tilde{H} | u_\alpha^{\mathbf{k}} \rangle}{(E_\alpha(\mathbf{k}) - E_\beta(\mathbf{k}))^2} \\
&= -i\frac{e^2}{\hbar} \sum_{E_\alpha < E_F < E_\beta} \int_{\mathbf{T}^2} \frac{d^2k}{(2\pi)^2} \langle \partial_y u_\alpha^{\mathbf{k}} | u_\beta^{\mathbf{k}} \rangle \langle u_\beta^{\mathbf{k}} | \partial_x u_\alpha^{\mathbf{k}} \rangle - \langle \partial_x u_\alpha^{\mathbf{k}} | u_\beta^{\mathbf{k}} \rangle \langle u_\beta^{\mathbf{k}} | \partial_y u_\alpha^{\mathbf{k}} \rangle \\
&= -i\frac{e^2}{\hbar} \sum_{\alpha} \int_{\mathbf{T}^2} \frac{d^2k}{(2\pi)^2} \langle \partial_y u_\alpha^{\mathbf{k}} | \partial_x u_\alpha^{\mathbf{k}} \rangle - \langle \partial_x u_\alpha^{\mathbf{k}} | \partial_y u_\alpha^{\mathbf{k}} \rangle
\end{aligned} \tag{4.31}$$

Compare the integrand with the definition of Berry curvature (4.28), we proof the formula (4.29). $\square$

Chern insulators The simplest case is following single-particle Hamiltonian (in the momentum space)

$$\tilde{H}(\mathbf{k}) = \vec{E}(\mathbf{k}) \cdot \vec{\sigma} + \epsilon(\mathbf{k})\mathbf{1} \tag{4.32}$$

There are only two bands, each with energy $\epsilon(\mathbf{k}) \pm E(\mathbf{k})$. For any model, we can introduce a unit vector on the two-sphere $\vec{n}(\mathbf{k}) = \frac{\vec{E}(\mathbf{k})}{|\vec{E}(\mathbf{k})|}$. Clearly this induces a map from $\mathbf{T}^2 \rightarrow \mathbf{S}^2$. Then the Chern number for this system could be written as

$$C = \frac{1}{4\pi} \int_{\mathbf{T}^2} k\vec{n} \cdot \left(\frac{\partial \vec{n}}{\partial k_x} \times \frac{\partial \vec{n}}{\partial k_y} \right) \tag{4.33}$$

This formula effectively measures how many times $\mathbf{T}^2$ wraps around $\mathbf{S}^2$.

A simple understanding of non-zero Chern number

Let us briefly summarize what we are studying here. For a system with a compact space of momentum, we have a map from the Brillouin zone to parameter space, that is

$$\phi : \mathbf{k} \mapsto \mathbf{d}(\mathbf{k}).$$

So if we want a non zero Chern number, each point on the two sphere must be taken at least once, which means $\mathbf{T}^2$ wraps $\mathbf{S}^2$.

Lattice Chern insulators Consider the symmetry allowed by crystal and the low energy effective physics around small momentum region, the Hamiltonian necessarily looks like

$$H = \sin(k_x)\sigma_x + \sin(k_y)\sigma_y + B(2 + M - \cos(k_x) - \cos(k_y))\sigma_z, \tag{4.34}$$

to the lowest order of Fourier modes.

5 Chern Insulators, Time Reversal Symmetry and Topological Insulators

I am not sure about the physical picture of this chapter. In a sense, I think that topological phase transition is not a good description of the main idea of this chapter.

Instead, we talk about how different mass term are related with topology of insulators. That is, symmetry of system usually constrains possible terms appearing in Hamiltonian, which are responsible for gap-closing in the low energy limit. So if a certain symmetry is required, a topological phase may be forbidden. For example, if time reversal symmetry is required, system cannot have non-zero Chern number generally, thus the Hall conductance is zero.

On the other hand, it can be shown that upon maintaining time reversal symmetry, we can still have other non-trivial topological phases that are categorized by $\mathbb{Z}_2$ symmetry. And they are achieved by adding different kinds of mass terms.

5.1 Time reversal symmetry

Generally, time reversal symmetry would protect systems from non-trivial topological phases. For spinless systems, time reversal symmetry acting on Bloch Hamiltonian as simply changing the sign of momentum

$$Th(\mathbf{k})T^{-1} = h(-\mathbf{k}) \quad (5.1)$$

Then, if $|\psi(\mathbf{k})\rangle$ is an eigenstate at momentum $\mathbf{k}$, $T|\psi(\mathbf{k})\rangle$ is an eigenstate at $-\mathbf{k}$.

5.2 Other kinds of symmetries

6 The Fractional Quantum Hall Effect

Given the knowledge of integer quantum Hall effect, we now come to an understanding of fractional quantum Hall effect. Indeed, the above argument being relevant with topology has a loophole and ultimately they hold only for non-interacting particles.

6.1 The Laughlin Wavefunction

The very first attempt made by physicist was Laughlin in 80' that he proposed an exact form of multi-particle wavefunction and obtained the fractional Landau level,

$$\nu = \frac{1}{m}, \quad m \in \mathbb{Z} \quad (6.1)$$

Actually, m is an odd integer due to the Fermi-Dirac statistics. As we see from solving the single-particle wavefunction at each Landau level, the wavefunction may take the following form when working in the symmetric gauge

$$\psi(z_1, \dots, z_n) = f(z_1, \dots, z_n) e^{-\sum_{i=1}^N |z_i|^2 / 4l_B^2} \quad (6.2)$$

where z_i is defined as the holomorphic coordinate for each particle. Laughlin proposed that the ground state wavefunction for Landau level $\nu = \frac{1}{m}$ is:

$$\psi(z_i) = \prod_{i < j} (z_i - z_j)^m e^{-\sum_{i=1}^N |z_i|^2 / 4l_B^2} \quad (6.3)$$

This is anti-symmetric when m is odd. For even m , this can be seen as the model for quantum Hall effect for bosons.

We can use some approximation to see that this function indeed give rise to a fractional Landau level. Let us focus on one single particle, say z_1 . Then in the pre-factor, terms containing z_1 are

$$\prod_{i=1}^N (z_1 - z_i)^m \sim z_1^{m(N-1)} \quad (6.4)$$

Thus from what we see from the single-particle wavefunction, particles now living within the maximum radius of $R \approx \sqrt{2mN}l_B$. And this implies an area of $A = \pi R^2 = 2\pi mNl_B^2$. Each Landau level contains $AB/\Phi_0 = mN$. So states only fill a fraction of Landau level with

$$\nu = \frac{1}{m} \quad (6.5)$$

as we expected.

It is quite surprising that as one solve the Shrödinger numerically, the Laughlin wavefunction matches with it perfectly. However, there is no reason for us to believe this would match when there are 10^{11} particles. Instead, we could think that this wavefuction lives in the same *universality class* as the true ground state.

Plasma analogy We would like to give another physical picture for the Laughlin wavefunction. Consider the number density in the ground state, which is given by computing the expectation value of

$$n(z) = \sum_{i=1}^N \delta(z - z_i) \quad (6.6)$$

And this is given by

$$\langle n(z) \rangle = \frac{\int \prod_{i=1}^N d^2 z_i n(z) |\psi(z_i)|^2}{\int \prod_{i=1}^N d^2 z_i |\psi(z_i)|^2} \quad (6.7)$$

Here we can interpret it as computing the partition function of a classical plasma, where the partition function is defined by

$$Z = \int \prod_{i=1}^N d^2 z_i |\psi(z_i)|^2 = \int \prod_{i=1}^N d^2 z_i \prod_{i<j} |z_i - z_j|^{2m} e^{-\sum_{i=1}^N |z_i|^2 / 2l_B^2} \quad (6.8)$$

However for future convenience and a better physical picture, we should introduce a factor for normalization to make the integrand dimensionless, that is

$$Z = \int \prod_{i=1}^N d^2 z_i \prod_{i<j} \left| \frac{z_i - z_j}{l_B} \right|^{2m} e^{-\sum_{i=1}^N |z_i|^2 / 2l_B^2} \quad (6.9)$$

In the usual computation of partition function, the integrand is given by $e^{-\beta H}$, so here we can identify the effective Hamiltonian as

$$\beta H = -2m \sum_{i<j} \log \left| \frac{z_i - z_j}{l_B} \right| + \sum_{i=1}^N \frac{|z_i|^2}{2l_B^2} \quad (6.10)$$

The first term can be interpreted as the Coulomb interaction between particles in 2d space, while the second term can be interpreted as a confining potential. If we set $\beta = \frac{2}{m}$, then the effective Hamiltonian is given by

$$H = -m^2 \sum_{i < j} \log \left| \frac{z_i - z_j}{l_B} \right| + \sum_{i=1}^N \frac{m}{4l_B^2} |z_i|^2 \quad (6.11)$$

where each single particle carries a charge m . One can easily check that the second term is the interaction of particles with a uniform background with charge density $\rho_B = -\frac{1}{2\pi l_B^2}$. To minimize the energy, particles would like to spread out in the space and neutralize the background charge. Thus we have a particle density of $n = \frac{1}{2\pi m l_B^2}$, which is exactly the same as what we have found before.

6.2 Excitation states: quasi-holes and quasi-particles

To this end, we have already discussed the ground state of $\nu = 1/m$ quantum Hall systems but not their excitations. Now we consider quasi-hole excitations and their wave functions, which are easily constructed. On the other hand, we would not discuss quasi-particle wave functions since they have not been exactly shown in the literature in a simple way.

The wave function for a quasi-hole excitation at position $\eta \in \mathbb{C}$ is given by

$$\psi(z_i) = \prod_{i=1}^N (z_i - \eta) \prod_{k < l} (z_k - z_l)^m e^{-\sum_{i=1}^N |z_i|^2 / 4l_B^2} \quad (6.12)$$

If there are M quasi-hole excitations at positions $\eta_1, \dots, \eta_M$, the wave function is given by

$$\psi(z_i) = \prod_{i=1}^N \prod_{j=1}^M (z_i - \eta_j) \prod_{k < l} (z_k - z_l)^m e^{-\sum_{i=1}^N |z_i|^2 / 4l_B^2} \quad (6.13)$$

A heuristic way to understand that the quasi-hole carries a charge of $\frac{e}{m}$ is that we could place m quasi-holes at the same position η , where we have defined that electrons carry a charge of $-e$. The wave function in this case is given by

$$\psi(z_i) = \prod_{i=1}^N (z_i - \eta)^m \prod_{k < l} (z_k - z_l)^m e^{-\sum_{i=1}^N |z_i|^2 / 4l_B^2} \quad (6.14)$$

which is exactly the same as the original Laughlin wave function with an additional electron at position η if we promote η to be a dynamical variable. However, η is just a parameter in (6.12) and (6.13), thus they have a fractional charge of $\frac{e}{m}$.

6.3 Topology behind fractional quantum Hall effect

In the first section 4, we have already relate the integer quantum Hall effect with its deeply hidden topology. A natural question is whether we can also find the topological nature of fractional quantum Hall effect. The two stories are quite different as we have already seen. But the answer should certainly be yes and we will give a brief discussion here.

Let us first discuss the statistics of quantum particles in 2d space. In the usual 3d space, we have two types of particles in quantum mechanics, bosons and fermions, which pick up a phase of 1 and -1 when we exchange two identical particles, respectively. However, in 2d space, we can have more general statistics, which is called *anyonic statistics*. That is, when we exchange two identical particles, the wave function would pick up a phase of $e^{i\theta}$ where θ can be any value in $[0, 2\pi)$.

Consider two identical particles rotate in two different directions. In the first case they exchange positions twice in an anti-clockwise direction. In the second case in a clockwise direction. If one inspect their worldlines, one would find that the two configurations could not be continuously deformed from one to the other. That is, although the modulus square is the same, the wave function may have a memory from what they experienced.

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = e^{i\theta} \psi(\mathbf{r}_2, \mathbf{r}_1) \quad (6.15)$$

where θ is not required to be 0 or π , instead it can be any value in $[0, 2\pi)$. This is so called *quantum statistics* or *fractional statistics*.

6.4 Further results

There are other fractions that are not of the form $\nu = \frac{1}{m}$ found in experiments. They can be understood in a similar manner from previous discussion.

7 Quantum Hall Effect From Chern-Simons Theory

In this section, I briefly summarize how can we get quantum Hall effect from a field theory view. We integrate over all degrees of freedom and are left with external gauge fields A_μ and emergent gauge fields a_μ . The lowest order allowed in $2 + 1d$ is given by Chern-Simons term.

7.1 Integer Quantum Hall Effect

We first investigate how to describe integer quantum Hall effect using Chern-Simons theory. It is assumed that we integrate over all internal degrees of freedom and only an effective action for A_μ is left. We assume that translation symmetry and rotation symmetry are still preserved, however parity and time reversal symmetry are both broken. As we are now in $2 + 1d$ spacetime, the lowest order term is given by Chern-Simons term

$$S_{CS} = \frac{k}{4\pi} \int d^3x A \wedge dA = \frac{k}{4\pi} \int d^3x \epsilon^{\mu\nu\rho} A_\mu \partial_\nu A_\rho \quad (7.1)$$

Then the electromagnetic current is given by varying the action with respect to the field,

$$J^\mu = \frac{\delta S}{\delta A_\mu} = \frac{k}{2\pi} \epsilon^{\mu\nu\rho} \partial_\nu A_\rho \quad (7.2)$$

where current conservation is satisfied automatically. The time component is given by

$$J^0 = \frac{k}{2\pi} \epsilon^{0\nu\rho} \partial_\nu A_\rho = \frac{k}{2\pi} B. \quad (7.3)$$

that is, if we add external magnetic field on the material, there would be induced charge from the magnetic field. At the same time, spatial components are computed by

$$J^i = \frac{k}{2\pi} \epsilon^{ij} E_j \quad (7.4)$$

and Hall conductivity is easily read from above equation

$$\sigma_{xy} = \frac{k}{2\pi} \quad (7.5)$$

Up to now we haven't shown why this is quantized, which needs a more careful study of self-consistence of Chern-Simons theory. Firstly, we want this theory to be gauge-invariant. Let $A_\mu \rightarrow A_\mu + \partial_\mu \omega$, then the variation of the action is

$$S_{\text{CS}} \rightarrow S_{\text{CS}} + \frac{k}{4\pi} \int d\omega \wedge dA = S_{\text{CS}} + \frac{k}{4\pi} \int d(\omega \wedge dA). \quad (7.6)$$

The boundary term vanishes if we put ourselves on a non-compact spacetime and consider transform near identity, which mean $\omega(x) \rightarrow 0$ as $x \rightarrow \infty$. However if this is considered to be a partition function with periodicity β , we may introduce *large gauge transformations*. Now ω is no longer required to be single-valued, we only want $e^{ie\omega/\hbar}$ to be single-valued as one move from $\tau = 0 \rightarrow \tau = \beta$. This implies a holonomy in S^1 time direction is possible

$$\omega = \frac{2\pi\hbar\tau}{e\beta}. \quad (7.7)$$

And A_0 can be thought as a periodic variable with periodicity $\frac{2\pi\hbar}{e\beta}$. Apart from time direction, we could also make the spatial directions compact, that is $\mathbf{S}^2$ instead of $\mathbf{R}^2$. And a background magnetic flux is imposed,

$$\frac{1}{2\pi} \int_{\mathbf{S}^2} dA = \frac{\hbar}{e} \quad (7.8)$$

If we assume a constant $A_0 = a$, then Chern-Simons action is calculated easily

$$S_{\text{CS}} = \frac{k}{2\pi} \int d^3x A_0 F_{12} = \frac{ka\beta\hbar}{e} \quad (7.9)$$

while $e^{iS_{\text{CS}}/\hbar}$ is required to be single-valued as $A_0 \rightarrow A_0 + \frac{2\pi\hbar}{\beta e}$, thus

$$\frac{k\hbar}{e^2} = \nu \in \mathbf{Z} \quad (7.10)$$

In all, the Hall conductivity is required to be quantized

$$\sigma_{xy} = \frac{e^2}{2\pi\hbar} \nu \quad (7.11)$$

7.2 Fractional Quantum Hall Effect

As we move from the first full filling Landau level to partial filling Landau level, there the low-energy physics should be treated more carefully. We can no longer assume all

bulk degrees of freedom are integrated out, however, some of them must be collected to form a new topological degree of freedom, that is, we have an emergent gauge field a_μ .

The goal is to write down the effective action $S_{\text{eff}}[a_\mu, A_\mu]$. And we want to couple those two different gauge fields. As what we did in the former subsection, following form of current is possible

$$J^\mu = \frac{1}{2\pi} \epsilon^{\mu\nu\rho} \partial_\nu a_\rho \quad (7.12)$$

Then if we include both the dynamics of emergent field and coupling with external field, the action is given by

$$S_{\text{eff}}[a_\mu, A_\mu] = \int d^3x \frac{1}{2\pi} A \wedge da - \frac{m}{4\pi} A \wedge dA \quad (7.13)$$